

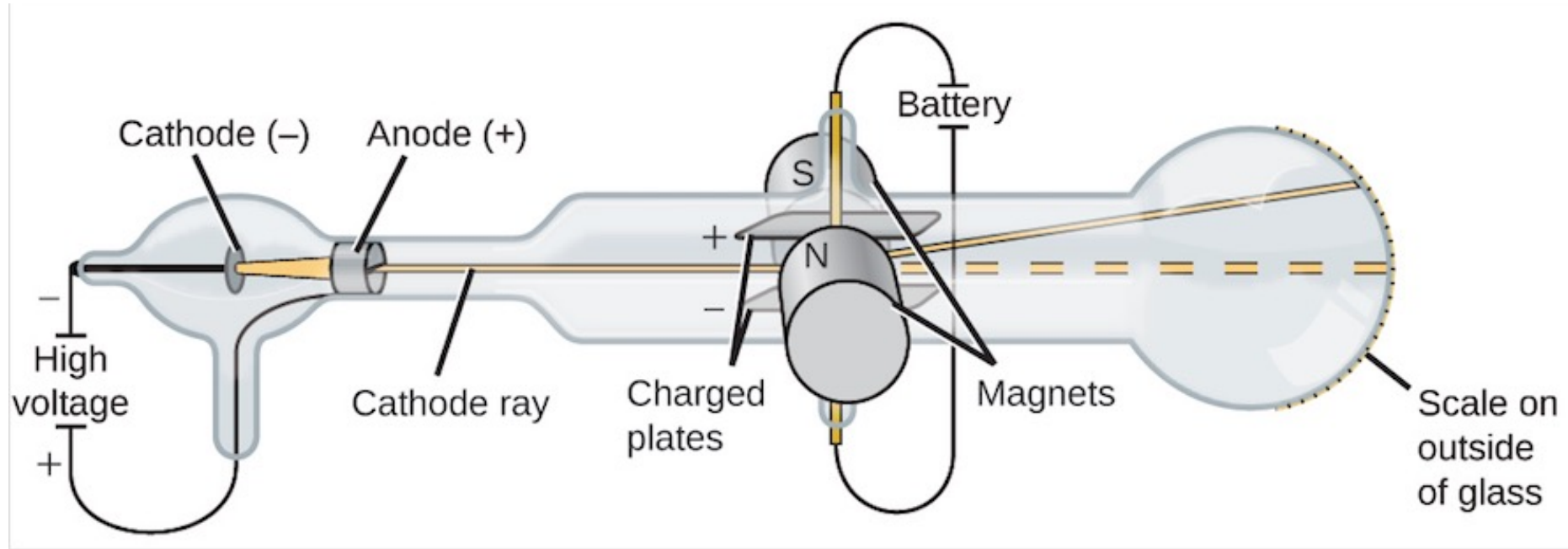
EC208_Unit 2_part 2

Dr. Dipjyoti Das

J.J. Thomson and the discovery of the electron

- ❑ Till 19th century, it was believed that all matter is made of indivisible particles called *atoms*, which cannot be created or destroyed.
- ❑ In the late 19th century, physicist J.J. Thomson began experimenting with *cathode ray tubes*.
- ❑ Cathode ray tubes are sealed glass tubes from which most of the air has been evacuated.
- ❑ A high voltage is applied across two electrodes at one end of the tube, which causes a beam of particles to flow from the cathode (the negatively-charged electrode) to the anode (the positively-charged electrode).
- ❑ The tubes are called cathode ray tubes because the particle beam or "cathode ray" originates at the cathode.
- ❑ The ray can be detected by painting a material known as phosphors onto the far end of the tube beyond the anode. The phosphors spark, or emit light, when impacted by the cathode ray.

J.J. Thomson and the discovery of the electron



- ❑ Thomson placed two oppositely-charged electric plates around the cathode ray. The cathode ray was deflected away from the negatively-charged electric plate and towards the positively-charged plate. This indicated that the cathode ray was composed of negatively-charged particles.
- ❑ Thomson also placed two magnets on either side of the tube and observed that this magnetic field also deflected the cathode ray. The results of these experiments helped Thomson determine the mass-to-charge ratio of the cathode ray particles.

J.J. Thomson and the discovery of the electron

❑ Thomson repeated his experiments using different metals as electrode materials and found that the properties of the cathode ray remained constant no matter what cathode material they originated from. From this evidence, Thomson made the following conclusions:

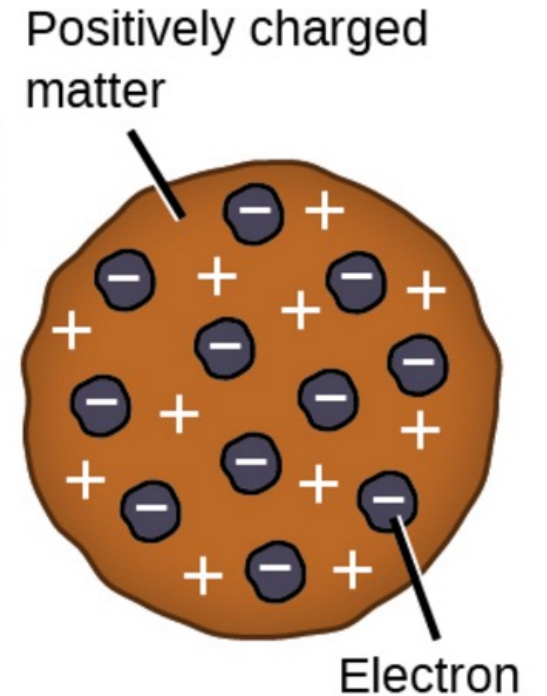
- The cathode ray is composed of negatively-charged particles.
- The particles must exist as part of the atom, since the mass of each particle is only $1/2000$ times the mass of a hydrogen atom.
- These subatomic particles can be found within atoms of all elements.

Eventually, his cathode ray particles were given a more familiar name: **electrons**.

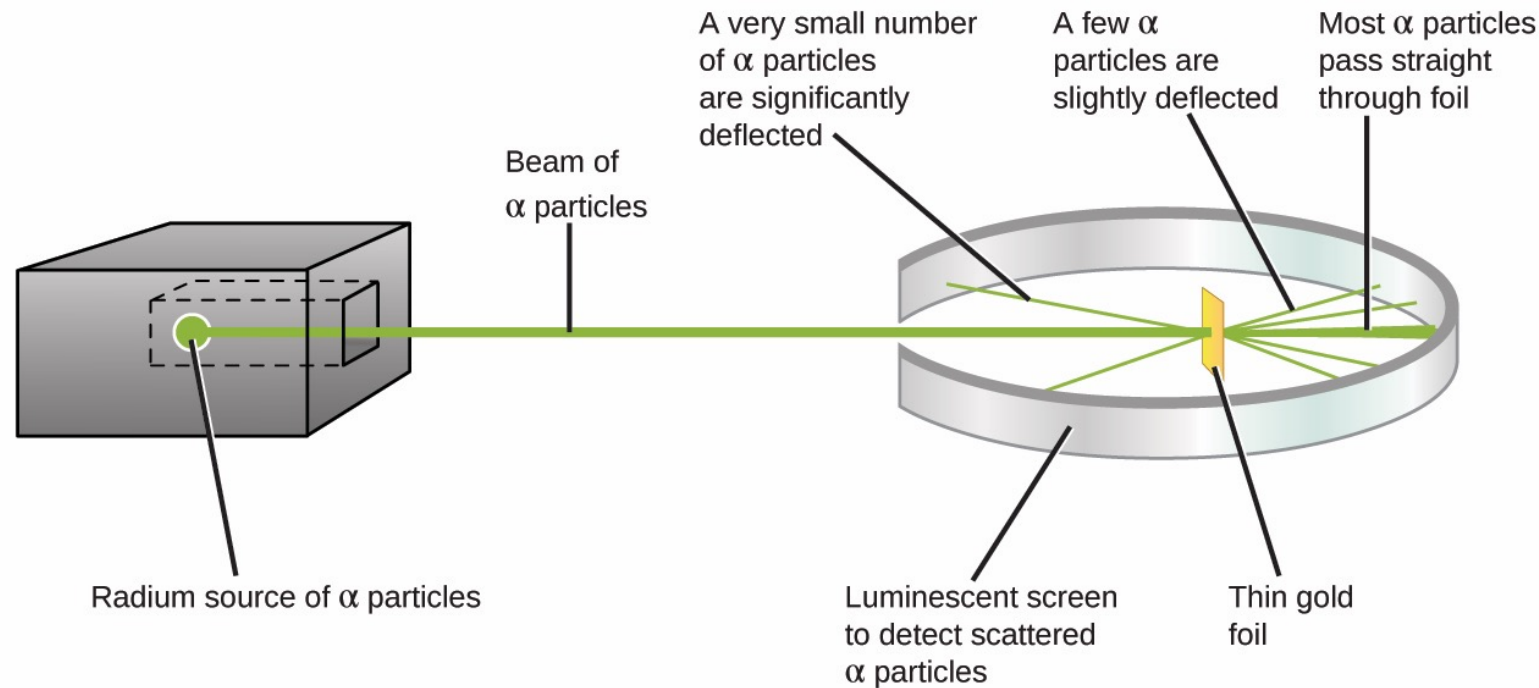
As part of his experiments with cathode ray tubes, Thomson tried changing the cathode material, which was the source of the particles. Since the same particles were emitted even when the cathode materials were changed to different metals, Thomson concluded that the particle was a fundamental part of all atoms.

Plum Pudding Model

- ❑ Thomson knew that atoms had an overall neutral charge. Therefore, he reasoned that there must be a source of positive charge within the atom to counterbalance the negative charge on the electrons.
- ❑ This led Thomson to propose that atoms could be described as negative particles floating within a soup of diffuse positive charge.
- ❑ This model is often called the *plum pudding model* of the atom, due to the fact that its description is very similar to plum pudding, a popular English dessert (see image below).



Ernest Rutherford and the gold foil experiment



- ❑ In his famous gold foil experiment, Rutherford fired a thin beam of α particles (pronounced *alpha* particles) at a very thin sheet of pure gold.
- ❑ Rutherford placed a sample of radium (a radioactive metal) inside a lead box with a small pinhole in it. Most of the radiation was absorbed by the lead, but a thin beam of α particles escaped out of the pinhole in the direction of the gold foil. The gold foil was surrounded by a detector screen that would flash when hit with an α particle.

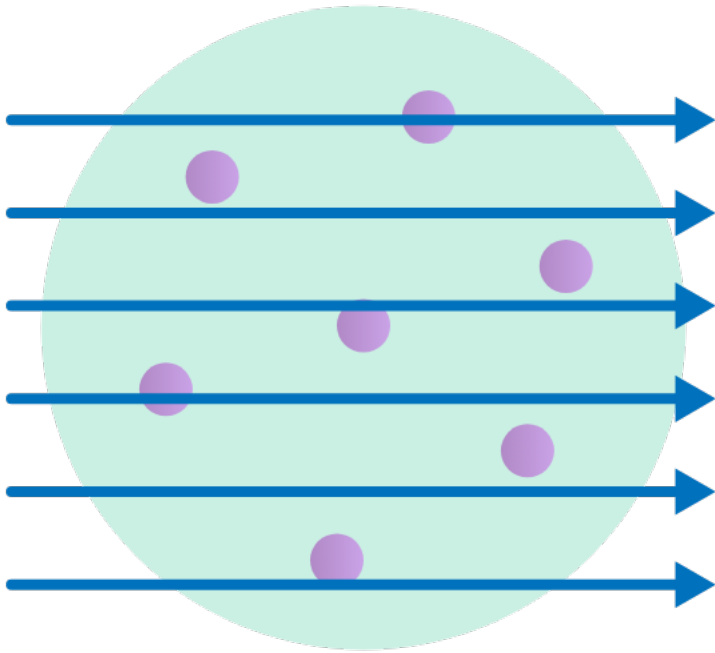
Ernest Rutherford and the gold foil experiment

- ❑ Rutherford predicted that most of the α particles would pass straight through the gold foil.
- ❑ The results of the experiment, however, were striking. While almost all of the α particles passed straight through the gold foil, a few α particles (about 1 in 20000) were deflected more than 90° from their path!
- ❑ Rutherford made the following conclusions about the structure of the atom:
 - The positive charge must be localized over a very tiny volume of the atom, which also contains most of the atom's mass. This explained how a very small fraction of the α particles were deflected drastically, presumably due to the rare collision with a gold nucleus.
 - Since most of the α particles passed straight through the gold foil, the atom must be made up of mostly empty space!

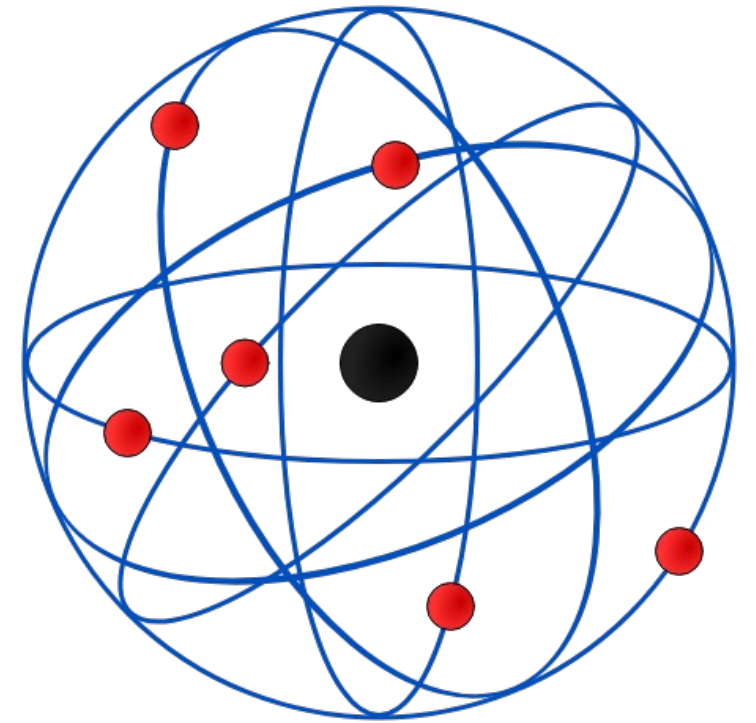
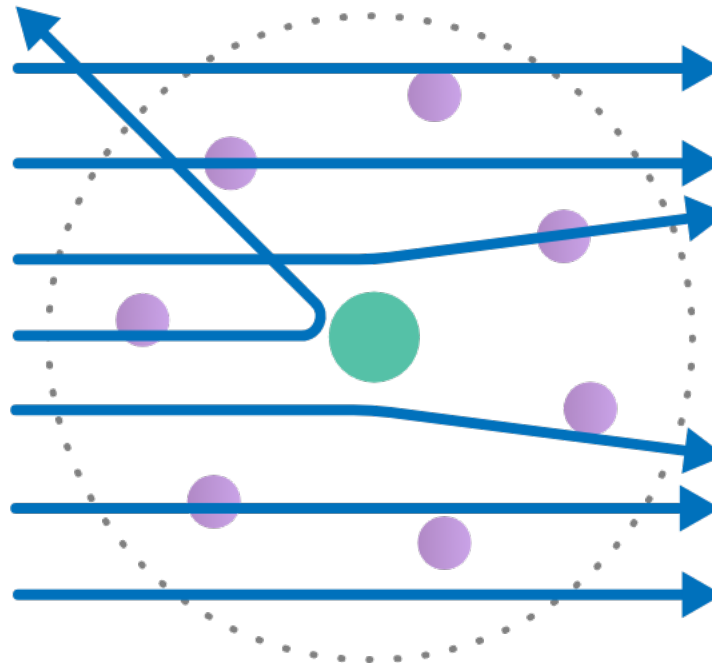
Ernest Rutherford and the gold foil experiment

- ❑ Rutherford proposed the nuclear model, in which an atom consists of a very small, positively charged nucleus surrounded by the negatively charged electrons.

THOMSON MODEL



RUTHERFORD MODEL



Rutherford atom dilemma

- ☐ Electrons revolve around the nucleus. How did the electrons keep themselves from collapsing into the nucleus, since opposite charges attract?
- ☐ Accelerated charge particle $\rightarrow$ e m radiation $\rightarrow$ energy loss $\rightarrow$ electron in nucleus?
- ☐ Was not able to explain discrete emission spectra of hydrogen.

Bohr's model

In 1913, Neils Bohr proposed a model that resolved both the Rutherford atom dilemma and discrete nature of spectra emitted by heated gas. Three postulates of Bohr's model:

- 1) The negative electrons move across the positive nucleus in a circular orbit. All electron orbits are centered at the nucleus. Not all classically possible orbits are available to an electron bound to the nucleus.
- 2) The allowed electron orbits satisfy the first quantization condition: In the n^{th} orbit, the angular momentum α_n of the electron can take only discrete values:

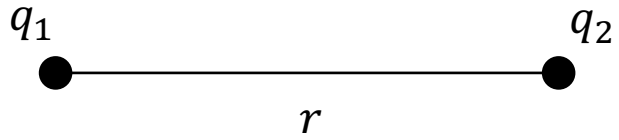
$$\alpha_n = \frac{nh}{2\pi}; n = 1, 2, 3, \dots$$

- 3) An electron is allowed to make transitions from one orbit where its energy is E_n to another orbit where its energy is E_m . When an atom absorbs a photon, the electron makes a transition to higher orbit when an atom emits a photon, the electron transits to a lower energy orbit.

The allowed energy transitions satisfy the second quantization condition:

$$h\nu = |E_n - E_m|$$

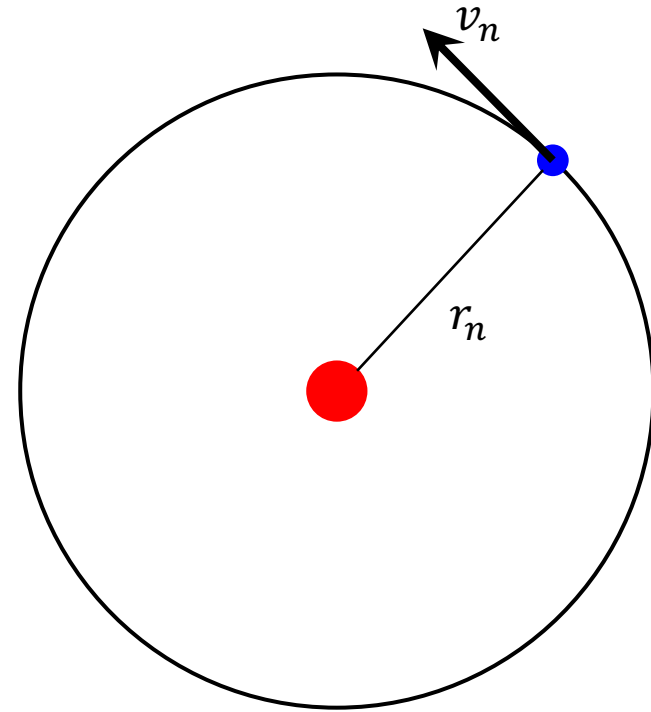
Derivation of Bohr Model



$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

$$PE = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r}$$

$$F_c = \frac{mv^2}{r}$$



Derivation of Bohr Model

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze(-e)}{r^2} = -\frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r^2}$$

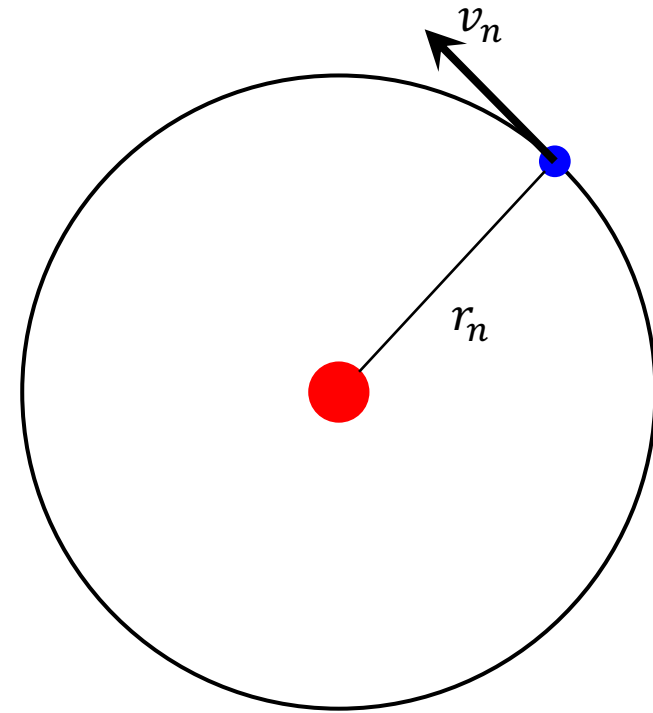
The negative direction indicates that the force is attraction.

$$\frac{mv_n^2}{r_n} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r_n^2}$$

For hydrogen atom, $Z=1$

$$\frac{mv_n^2}{r_n} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r_n^2}$$

$$mv_n^2 r_n = \frac{e^2}{4\pi\epsilon_0} \rightarrow (1)$$



Also, as per Bohr's postulate

$$L_n = mv_n r_n = \frac{nh}{2\pi}$$

$$mv_n r_n = n\hbar \rightarrow (2) \quad \hbar = \frac{h}{2\pi}$$

Derivation of Bohr Model

$$mv_n^2 r_n = \frac{e^2}{4\pi\epsilon_0} \rightarrow (1)$$

$$mv_n r_n = n\hbar \rightarrow (2)$$

Dividing (1) by (2), we get,

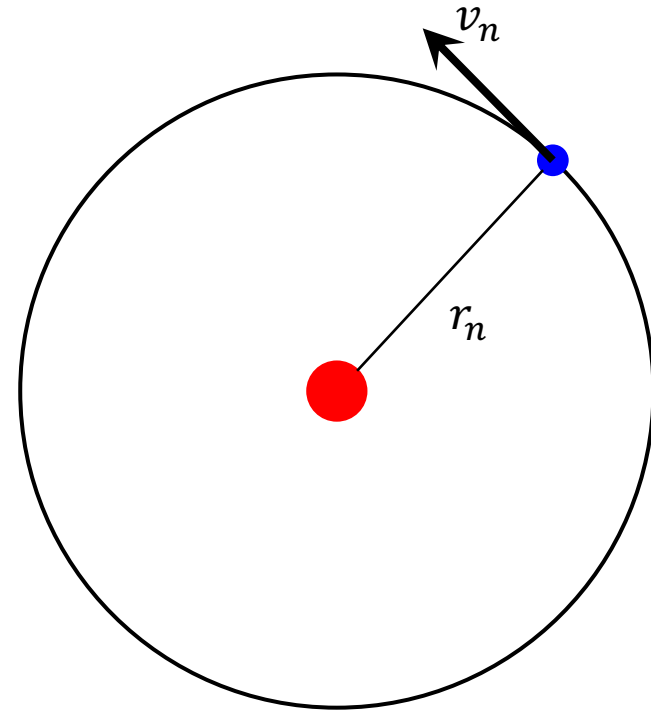
$$v_n = \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{\hbar} \cdot \frac{1}{n} \rightarrow (3)$$

$$r_n = \frac{n\hbar}{mv_n} = \frac{n\hbar}{m} \cdot \frac{4\pi\epsilon_0 n\hbar}{e^2}$$

$$r_n = 4\pi\epsilon_0 \frac{n^2 \hbar^2}{me^2} \rightarrow (4)$$

If we put $n=1$, we get Bohr radius

$$r_{bohr} = 4\pi\epsilon_0 \frac{\hbar^2}{me^2} = 0.529 \text{ \AA}$$



$$r_n \propto n^2$$

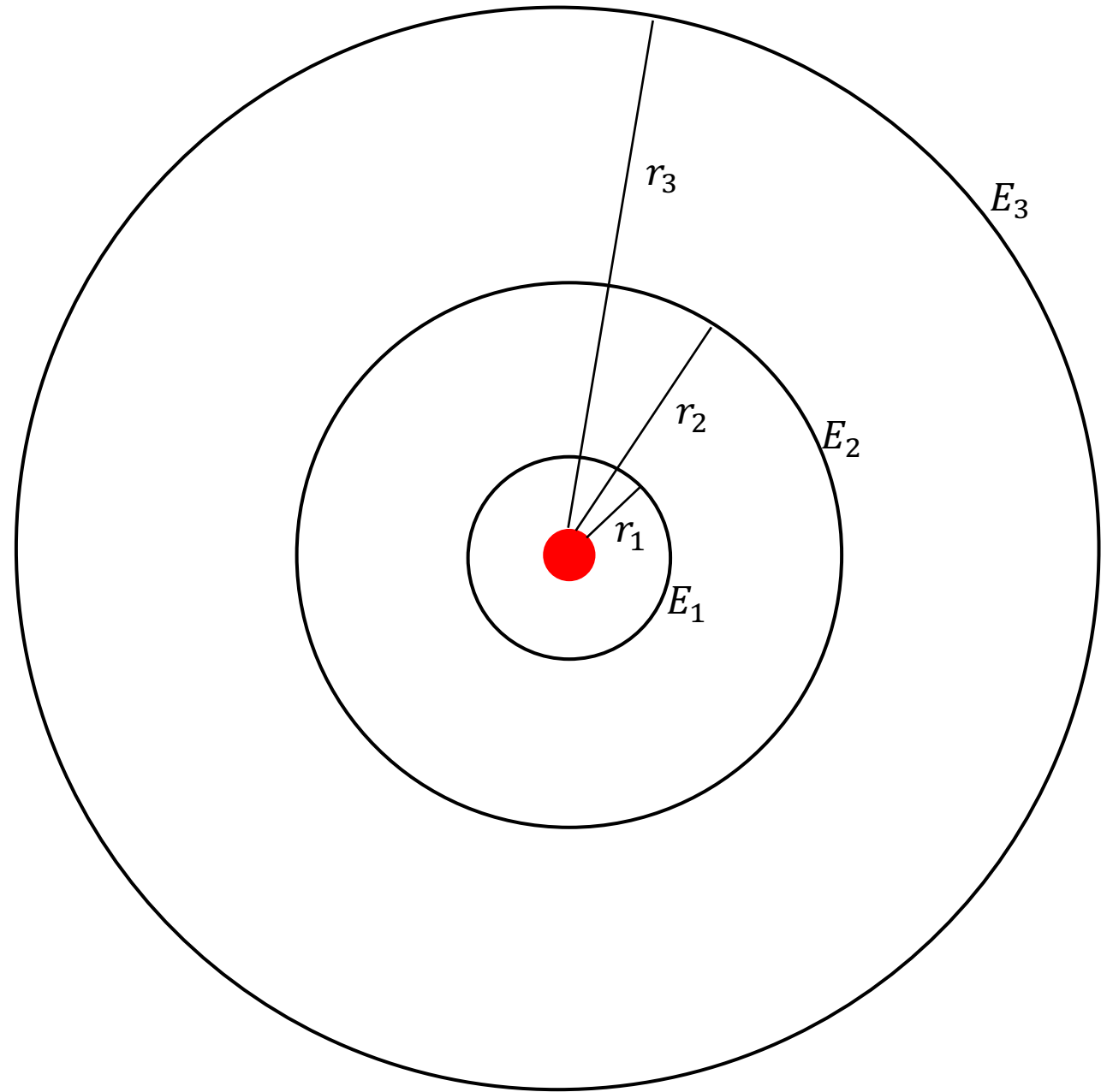
Derivation of Bohr Model

$$r_1 = 0.529 \text{ \AA}$$

$$r_2 = 2.166 \text{ \AA}$$

$$r_3 = 4.761 \text{ \AA}$$

$$E_n = ?$$



Derivation of Bohr Model

$$E_n = K_n (K.E.) + U_n (P.E.)$$

$$K_n = \frac{1}{2} m v_n^2 = \frac{1}{2} m \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{\hbar} \cdot \frac{1}{n} \right)^2$$

$$K_n = \frac{1}{32\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2}$$

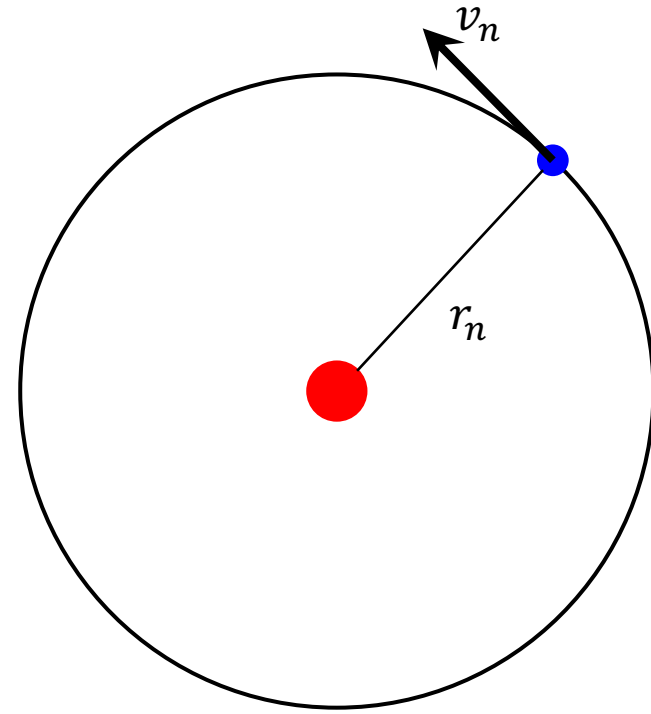
$$U_n = \frac{1}{4\pi\epsilon_0} \cdot \frac{-e^2}{r_n}$$

$$U_n = -\frac{1}{16\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2}$$

$$E_n = \frac{1}{32\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2} - \frac{1}{16\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2}$$

$$E_n = -\frac{1}{32\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2} = -E_0 \frac{1}{n^2}$$

$$E_0 = -13.6 \text{ eV}$$

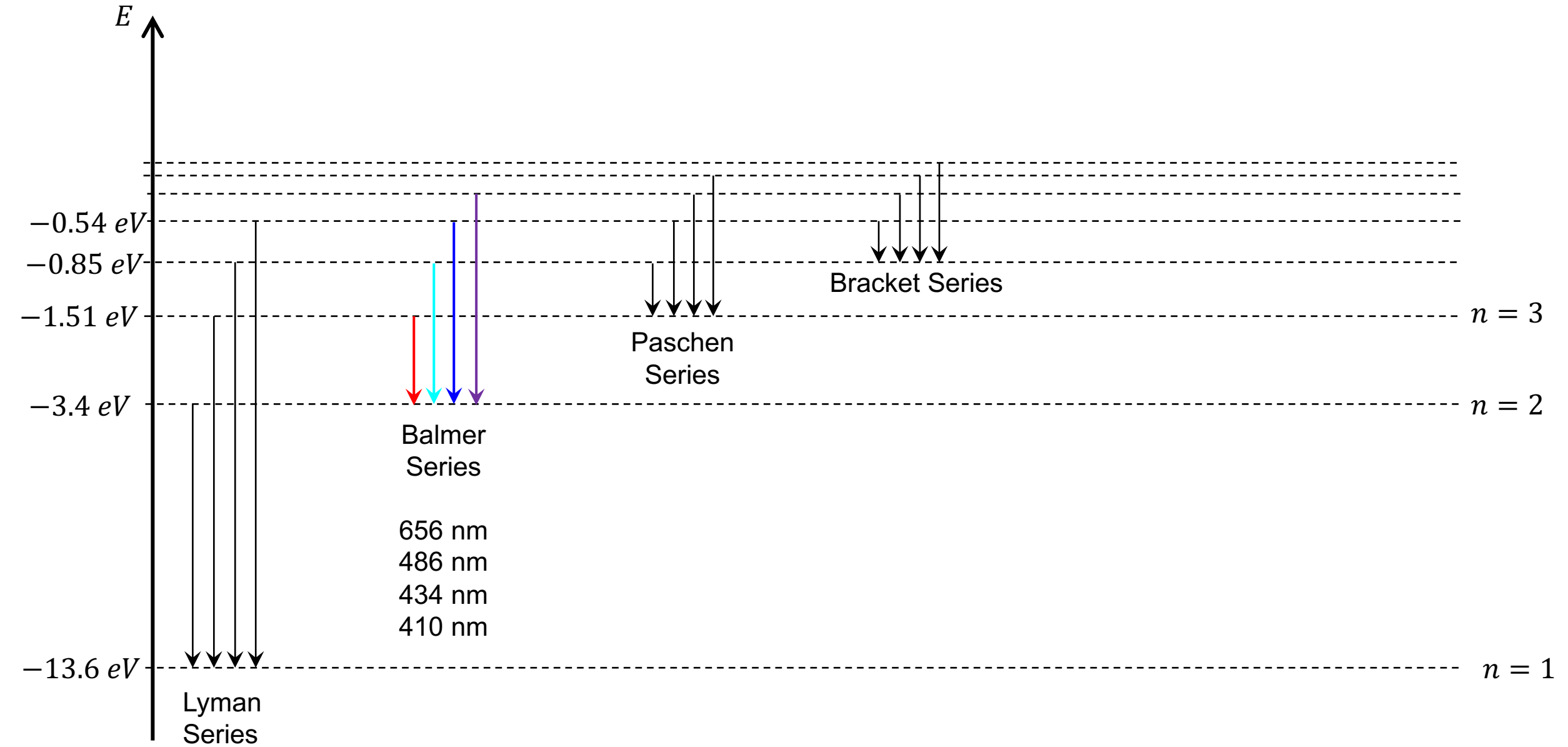


$$v_n = \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{\hbar} \cdot \frac{1}{n} \rightarrow (3)$$

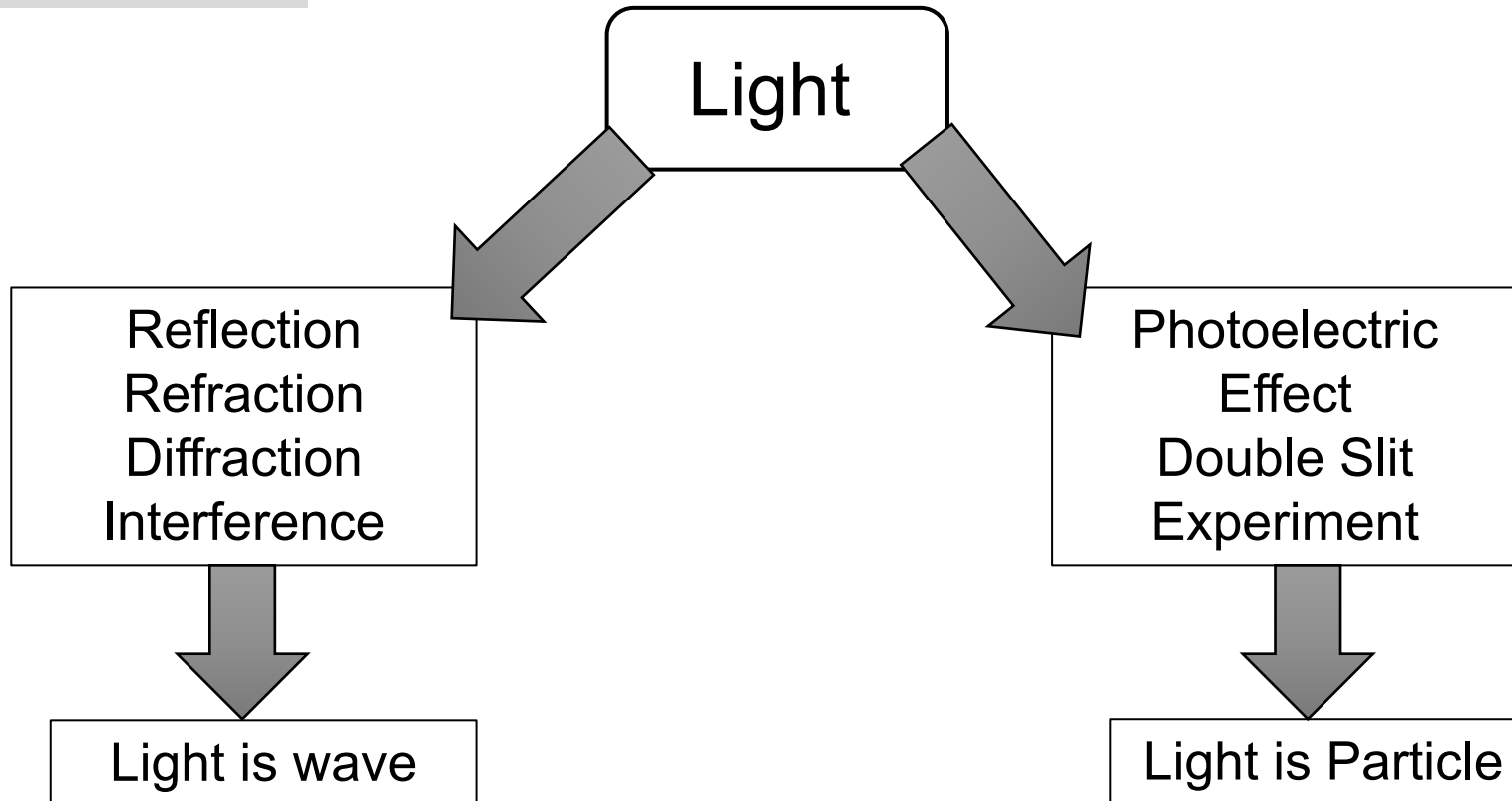
$$r_n = 4\pi\epsilon_0 \frac{n^2 \hbar^2}{me^2} \rightarrow (4)$$

Derivation of Bohr Model

$$E_n = -13.6 \frac{1}{n^2} \text{ eV}$$



Wave-Particle Duality



$c = \text{vel of light}$
 $p = \text{momentum}$

wavelength, λ

Frequency, f

$$c = \lambda f$$

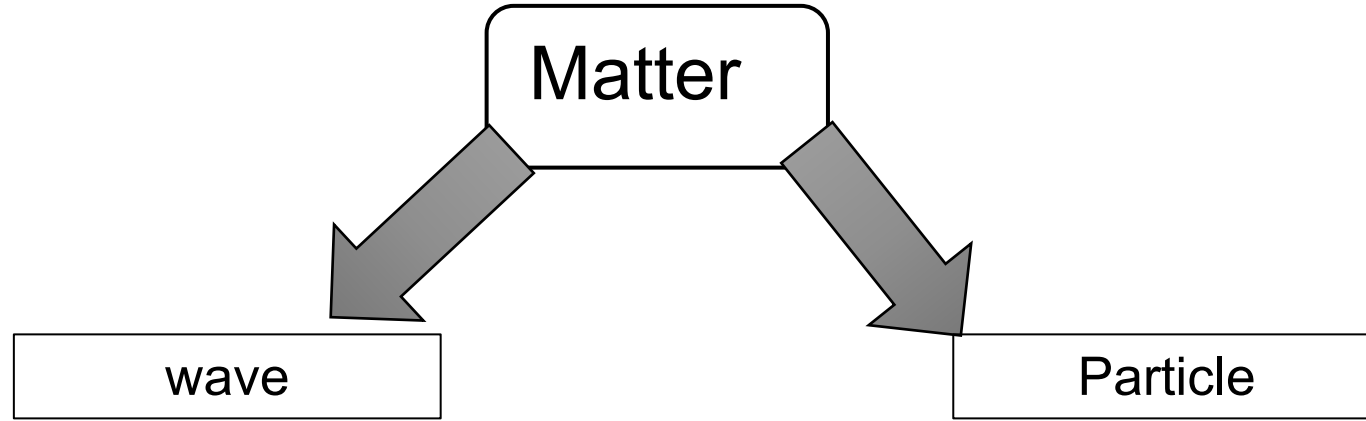
mass, m

velocity, v

$$p = ?$$

$$p = \frac{h}{\lambda}$$

De Broglie



Wave-Particle Duality: It is the concept in quantum mechanics that every particle exhibits both wave-like and particle-like properties. For example, electrons show diffraction (wave nature) as well as localized impacts in detectors (particle nature).

De Broglie Hypothesis: In 1924, Louis de Broglie proposed a new speculative hypothesis that electrons and other particles of matter can behave like waves.

Particles, like electrons, have an associated wavelength given by $\lambda = h/p$, where h is Planck's constant and p is the particle's momentum.

Wave Particle Duality



a tennis ball also
can be a wave?

Wave Particle Duality

Yes, according to de Broglie's hypothesis, every moving object has a wavelength given by $\lambda = h/p$ (where h is Planck's constant and p is momentum).

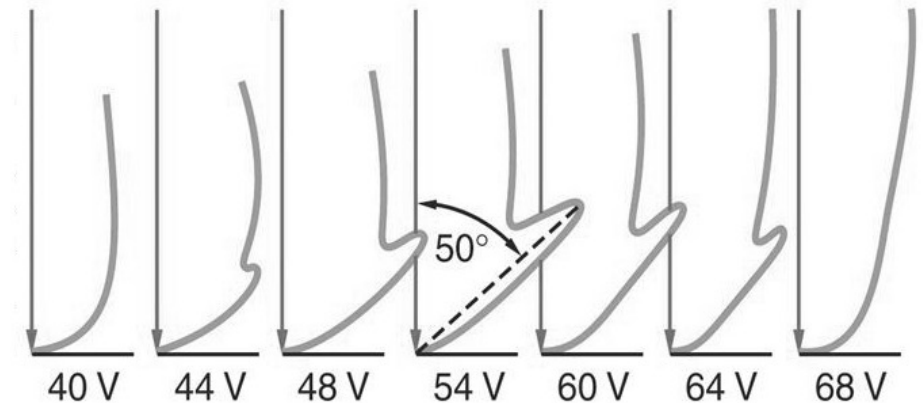
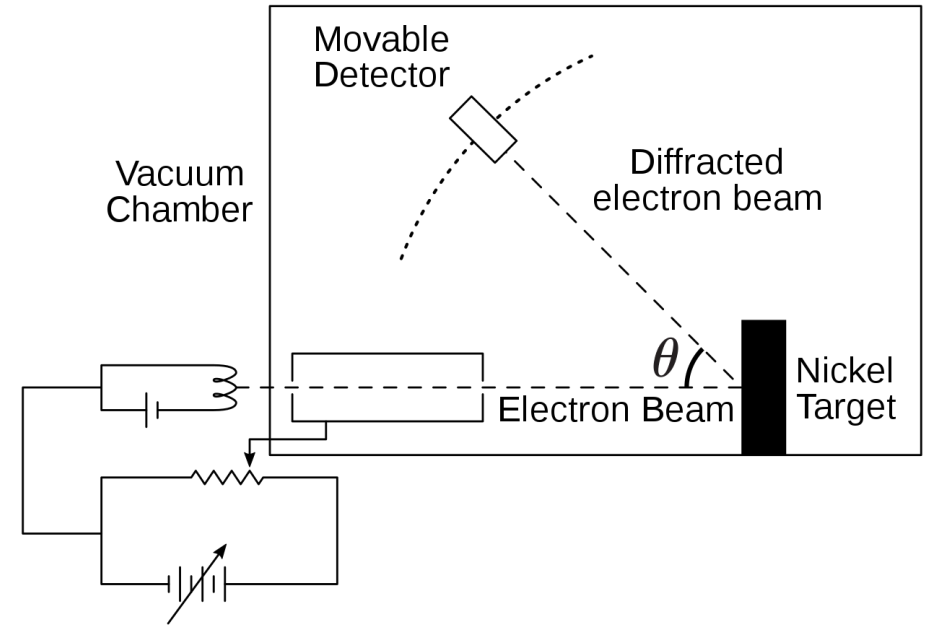
For macroscopic objects like a tennis ball, the wavelength is extremely small and practically undetectable.

If a 150 g tennis ball moves at 30 m/s, calculate its de Broglie wavelength $\lambda \approx 1.47 \times 10^{-34} \text{ m}$

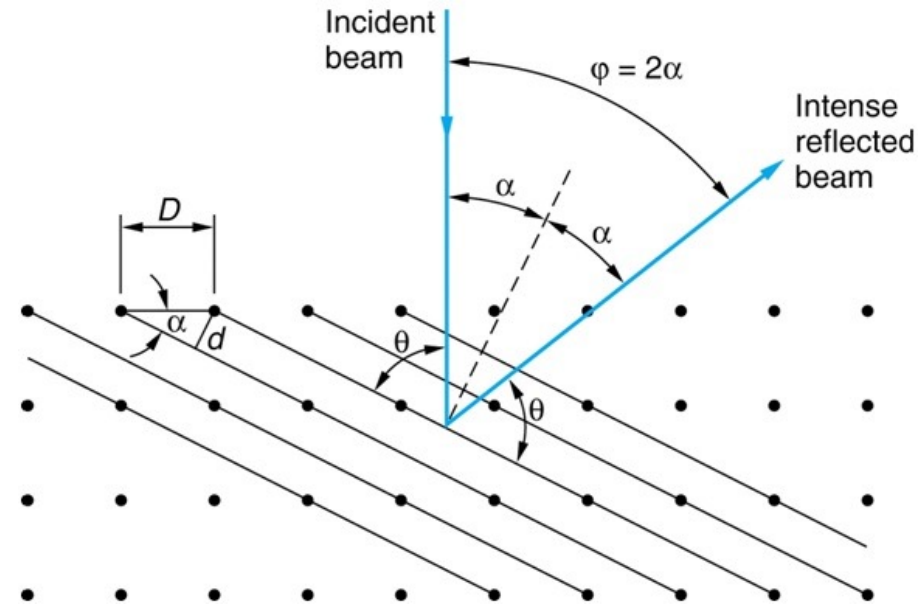
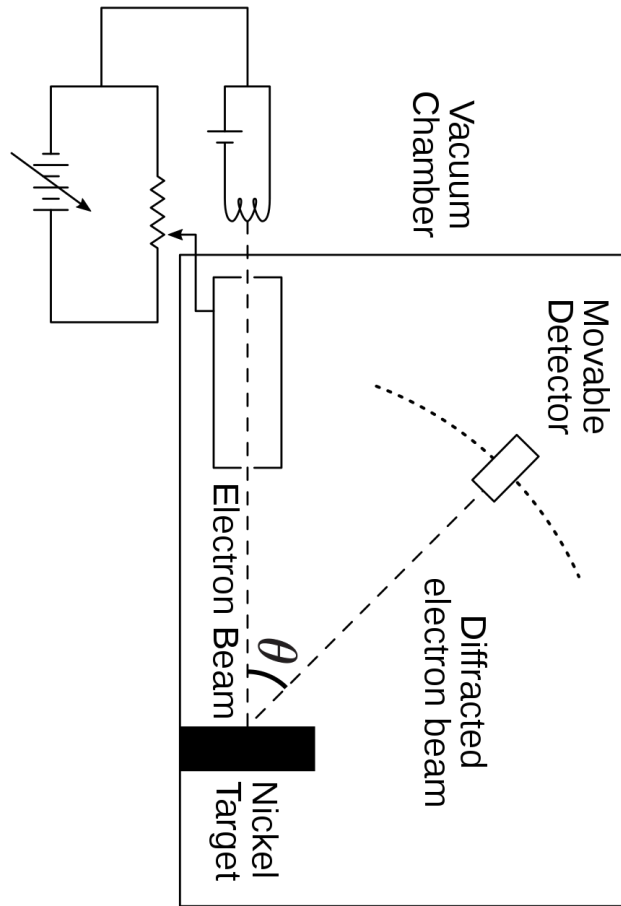
This is much smaller than atomic scales, making wave behavior irrelevant for large objects.

Davisson Germer Experiment

- ❑ Davisson and Germer Experiment, for the first time, proved the wave nature of electrons and verified the de Broglie equation.
- ❑ From the Davisson and Germer experiment, we get a value for the scattering angle θ and a corresponding value of the potential difference V at which the scattering of electrons is maximum.
- ❑ By varying the applied voltage to the electron gun, the maximum intensity of electrons diffracted by the atomic surface was found at different angles.
- ❑ The highest intensity was observed at an angle $\theta = 50^\circ$ with a voltage of 54 V, giving the electrons a kinetic energy of 54 eV.



Davisson Germer Experiment



$$n\lambda = 2d \sin \theta$$

$$\theta = 90 - 25 = 65^\circ$$

$$d = 0.091 \text{ nm}$$

$$\lambda = 0.166 \text{ nm}$$

De-Broglie

$$\text{wavelength, } \lambda = \frac{h}{p}$$

Energy of electron $qV = 54 \text{ eV} = 54 \times 1.6 \times 10^{-19} \text{ J}$

$$n\lambda = 2d \sin \theta$$

$$\theta = 90 - 25 = 65^\circ$$

$$d = 0.091 \text{ nm}$$

$$\lambda = 0.166 \text{ nm}$$

$$\text{Also, KE, } E = \frac{1}{2}mv^2 \Rightarrow mE = \frac{1}{2}(mv)^2$$

$$2mE = p^2$$

$$p = \sqrt{2mE} = \sqrt{2 * 9.1 * 10^{-31} * 54 * 1.6 * 10^{-19}} = 3.965 \times 10^{-24}$$

$$\text{wavelength, } \lambda = \frac{6.626 * 10^{-34}}{3.965 * 10^{-24}} = 0.167 \text{ nm}$$

Q. Find the KE of an electron having de-Broglie wavelength of 10 nm

Q. Find the KE of an electron having de-Broglie wavelength of 10 nm

$$\lambda = \frac{h}{p} = 10 * 10^{-9} = 10^{-8}$$

$$p = \frac{h}{10^{-8}} = 6.626 * 10^{-26}$$

$$p = \sqrt{2mE}$$

$$p^2 = 2mE$$

$$\frac{p^2}{2m} = E$$

$$E = \frac{(6.626 * 10^{-26})^2}{2 * 9.1 * 10^{-31}} = 2.41 * 10^{-21} J$$

$$E = \frac{2.41 * 10^{-21}}{1.6 * 10^{-19}} = 0.015 eV$$